**10 (a)** Describe how the piezoelectric crystal in a transducer generates ultrasound waves for use in medical diagnosis.

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..... [3]

**(b)** A parallel ultrasound beam is incident on the boundary between two media, as illustrated in Fig. 10.1.

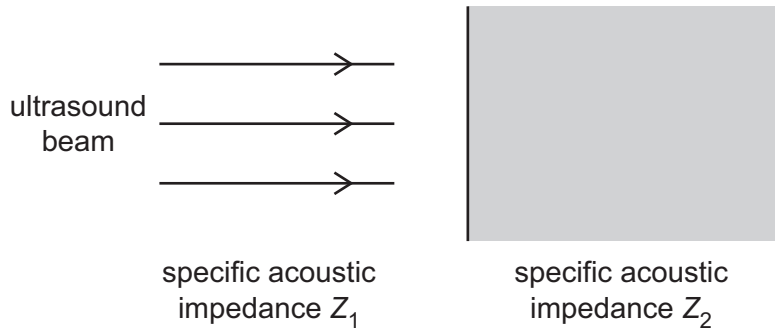



**Fig. 10.1**

The media have specific acoustic impedances $Z_1$ and $Z_2$.

At the boundary, a fraction $\alpha$ of the incident intensity of the ultrasound beam is reflected. The remainder is transmitted.

**(i)** State what is meant by specific acoustic impedance.

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..... [2]

**(ii)** Describe how $\alpha$ depends on the relative values of $Z_1$ and $Z_2$.

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**(c)** A parallel ultrasound beam of intensity $I_0$ enters a region of soft tissue. After passing a distance of 2.1 cm through this tissue, the intensity of the ultrasound is $0.62\,I_0$.

Calculate the linear attenuation coefficient $\mu$ of ultrasound in the soft tissue. Give a unit with your answer.

$\mu$ = ..... unit ..... [2]

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